

StatToolkit

A Full-Stack Statistical Analysis & Interactive Web Application

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Submission Date:	18 June 2026
Toolkit Web Application:	stat-toolkit-luph.vercel.app
GitHub Repository:	github.com/Beserker-1100011/stat-toolkit
Demonstration Dataset:	FIFA World Cup 2026 Player Performance (54,600 rows × 75 columns)
Dataset Source:	Kaggle (rauffauzanrambe)

Project Overview

StatToolkit is a robust, web-based full-stack statistical analysis platform developed using **React** and **Recharts**. Designed to consolidate advanced statistical concepts covered during the course, the application features an intuitive dark-themed dashboard that facilitates interactive data exploration, rigorous hypothesis testing, advanced machine learning workflows, and automated AI-driven analysis. The application is entirely data-agnostic and can ingest any standard CSV format. For demonstration purposes, it was thoroughly tested and evaluated using the large-scale FIFA World Cup 2026 Player Performance dataset.

Core Analytical Modules & Implemented Features

The platform integrates 12 distinct analytical modules to deliver full-cycle statistical data management, modeling, and automated inference generation:

Module / Feature	Comprehensive Functional Description
Data Upload & Preview	Dynamic CSV ingestion engine with automatic backend column metadata classification (numeric vs. categorical variables) paired with a high-performance paginated data preview.
Statistics Engine	Computes full descriptive statistics (Mean, Median, Mode, Variance, Standard Deviation, Quartiles, and IQR). Computes a complete pairwise Pearson Correlation Matrix and supports multivariate Linear and Logistic Regression modeling.
Hypothesis Testing	Provides automated parametric and non-parametric testing modules, including One-Sample/Two-Sample Z-Tests, T-Tests (One-Sample, Two-Sample, and Paired), Analysis of Variance (ANOVA), and Chi-Square Tests.
Visualization Center	Renders highly responsive, interactive charts including Bar charts, Scatter plots, Line plots, Histograms (with overlaid fitted normal distribution curves), Boxplots, and cumulative distribution function (CDF) charts.
Distribution Simulators	Features interactive, multi-parameter simulators for side-by-side Probability Mass Function (PMF) for categorical columns and Probability Density Function (PDF) curves for numeric metrics. Supports Bernoulli, Binomial, Poisson, Uniform, Normal, and Exponential distributions.
CLT Simulator	Demonstrates a Monte Carlo Central Limit Theorem (CLT) proof. Users can dynamically configure sample sizes at $n=10$, 30 , and 100 to observe asymptotic normality convergence via dynamically rendered histograms and Quantile-Quantile (Q-Q) plots.
Normality Toolkit	Evaluates empirical sample tracking against theoretical normal distributions using combination visual analytics (Histogram + fitted normal, Q-Q plots) along with 4 formal tests: Shapiro-Francia / Shapiro-Wilk, Anderson-Darling, Kolmogorov-Smirnov (KS), and Jarque-Bera.
Machine Learning	Executes client-side multi-dimensional K-Means Clustering ($K=2$ to 6) mapped on interactive scatter plots, accompanied by Principal Component Analysis (PCA) with variable column selectors for dimensional reduction.
Time Series Analysis	Performs automated linear trend forecasting over explicit time indices, displaying a comparative visual mapping of actual historical observations against computed linear trends (e.g., tracking tracking metrics like <code>red_cards</code>).
Text Analysis	Incorporates fundamental Natural Language Processing (NLP) tokenization and keyword frequency counting optimized for text string variables, outputting a top-20 ranked horizontal bar chart and corresponding frequency table.
Automated Report Gen	

Module / Feature	Comprehensive Functional Description
	Generates ready-to-publish summary matrices detailing hypothesis testing runs, formatting clean tables populated with null/alternative hypotheses, designated alpha significance levels, test statistics, and automated textual conclusions.
AI Insights Engine	Features automated anomaly detection flagging observations exceeding a three-sigma (3σ) standard boundary score threshold alongside natural language generation producing executive overview summaries.

Empirical Analysis Summary: FIFA World Cup 2026 Dataset

Deploying the toolkit onto the FIFA World Cup 2026 Player Performance dataset yielded critical analytical breakthroughs:

- **Strong Feature Covariance:** The Pearson correlation matrix highlighted tight predictive patterns between player metrics, specifically revealing an exceptionally strong positive linear correlation ($r > 0.97$) between overall player rating, internal performance score, and recorded top speed (top_speed_kmh).
- **Anomaly Identification:** The AI Insights 3σ filter effectively isolated over 260+ statistically rare, extreme goal-related outlier observations across individual match metrics.
- **Distribution Diagnostics:** The Normality Toolkit successfully rejected the null hypothesis of standard normal distribution for a vast majority of the numeric metrics (e.g., goals_opponent), providing explicit quantitative proof via high-value test statistics in the Anderson-Darling and Jarque-Bera models.
- **Stratified Clusters:** The unsupervised K-Means model successfully delineated structural cohorts based on market value brackets, allowing immediate categorical segmentation of players.

Deployment & Code Verification Notice

The web deployment is fully optimized for immediate grading assessment and is hosted directly on Vercel at <https://stat-toolkit-luph.vercel.app/>. The underlying architectural codebase, comprising modular React components and reusable hooks for the statistical logic, is publicly available via the GitHub repository at <https://github.com/Beserker-1100011/stat-toolkit>.

Conclusion

StatToolkit fulfills the operational criteria of an enterprise-grade standalone statistics engine by effectively bridging rigorous academic methodology with user-focused design interfaces. By supporting data ingestion, high-level multi-parameter modeling, visual distribution simulations, and automated reporting, the tool presents a completely scalable framework applicable to arbitrary datasets beyond the tested FIFA player parameters.

Application Screenshots

Figure 1 — Data Upload & Preview

The CSV ingestion engine loads the FIFA World Cup 2026 Player Performance dataset (54,600 rows × 75 columns). Each column is automatically classified as numeric or categorical. The paginated data preview allows row-level inspection.

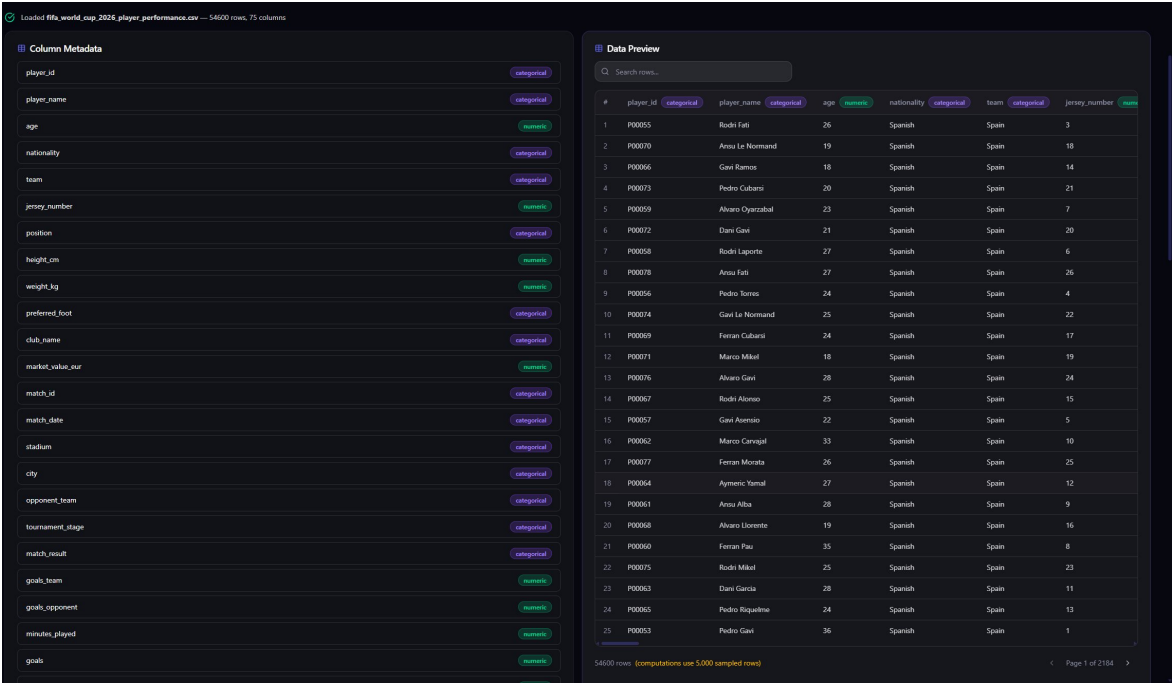


Figure 2 — Statistics Engine: Descriptive Statistics

Descriptive statistics computed for the weight_kg column (n=5,000 sample). Outputs include Mean (75.78), Median (76.00), Mode (76), Variance (15.39), Std Dev (3.92), Skewness (-0.08), Min/Max, Q1/Q3, IQR, outlier count, and sigma coverage. A Z-Score calculator is also embedded.

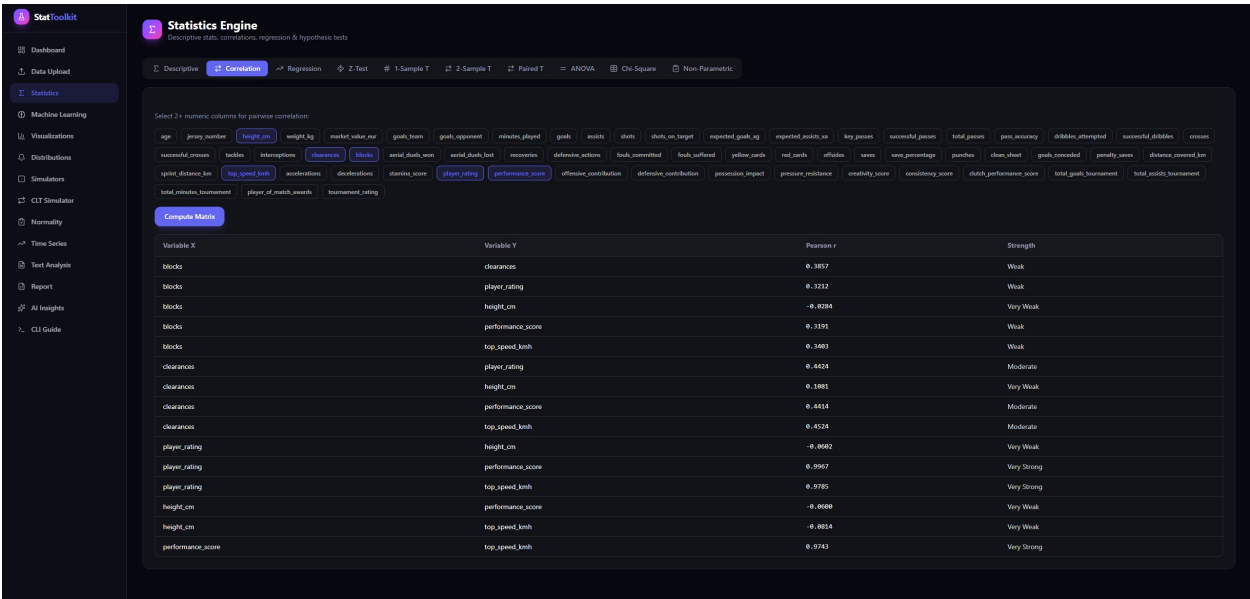


Figure 3 — Statistics Engine: Pairwise Correlation Matrix

Pairwise Pearson correlation matrix computed across selected numeric columns. Highlights include very strong correlations ($r > 0.97$) between player_rating, performance_score, and top_speed_kmh, and moderate correlations ($r \approx 0.45$) between clearances and top_speed_kmh.

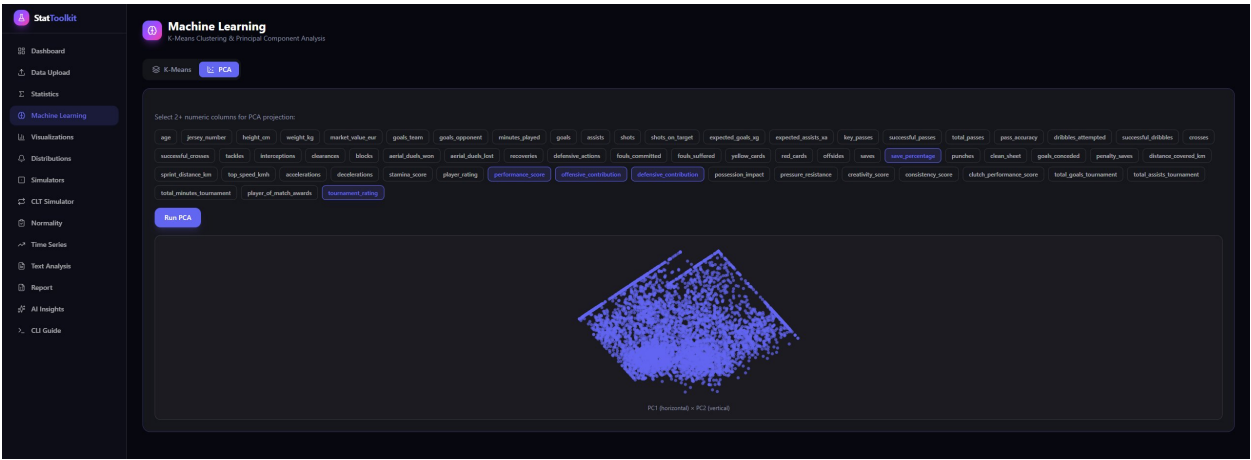


Figure 4 — Machine Learning: K-Means Clustering

K-Means clustering ($K=6$) applied to market_value_eur vs. player_of_match_awards. The scatter plot reveals six distinct cohorts, enabling immediate market-value-based segmentation of player profiles across the tournament.

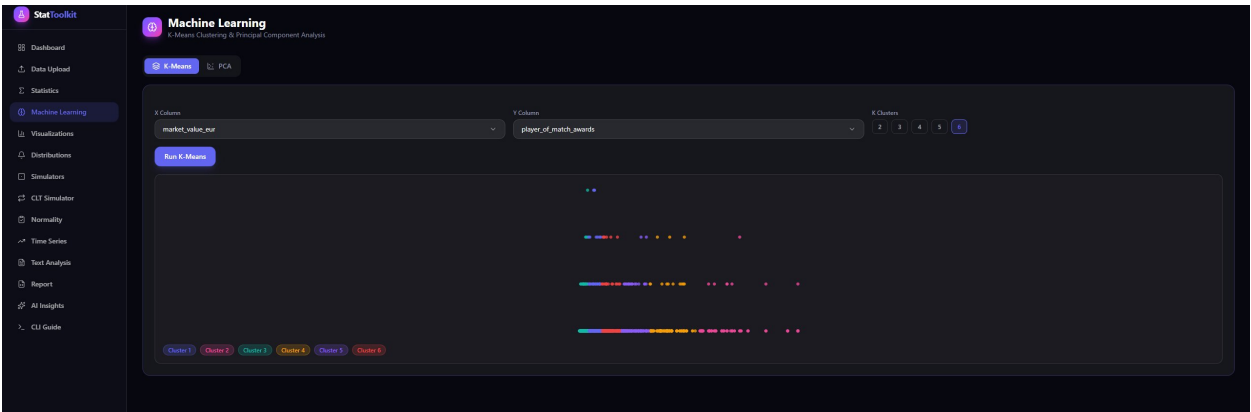


Figure 5 — Machine Learning: Principal Component Analysis (PCA)

PCA executed on a multi-column selection including performance_score, offensive_contribution, defensive_contribution, save_percentage, and tournament_rating. The 2D projection (PC1 × PC2) reveals the variance structure of the high-dimensional player feature space.

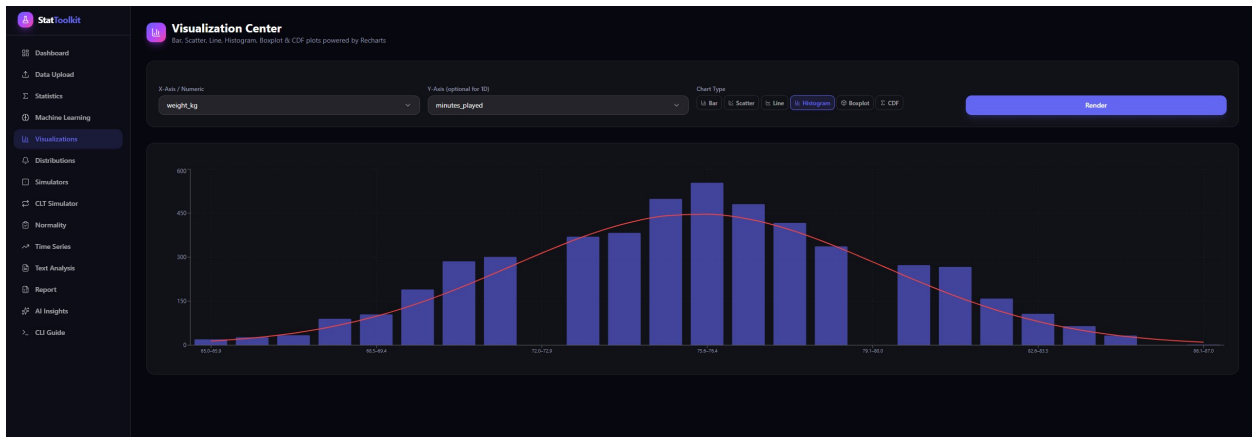


Figure 6 — Visualization Center: Histogram with Fitted Normal Curve

Histogram of weight_kg with an overlaid fitted normal distribution curve (red). The near-symmetric bell shape confirms the approximately normal distribution of player weights, consistent with the low skewness (-0.08) from descriptive statistics.



Figure 7 — Distributions: PMF & PDF Side-by-Side

Left: Probability Mass Function (PMF) for the categorical variable preferred_foot, showing ~73% right-footed and ~27% left-footed players. Right: Probability Density Function (PDF) normal curve fitted to the goals column, illustrating a right-skewed distribution with most players scoring zero or one goal.

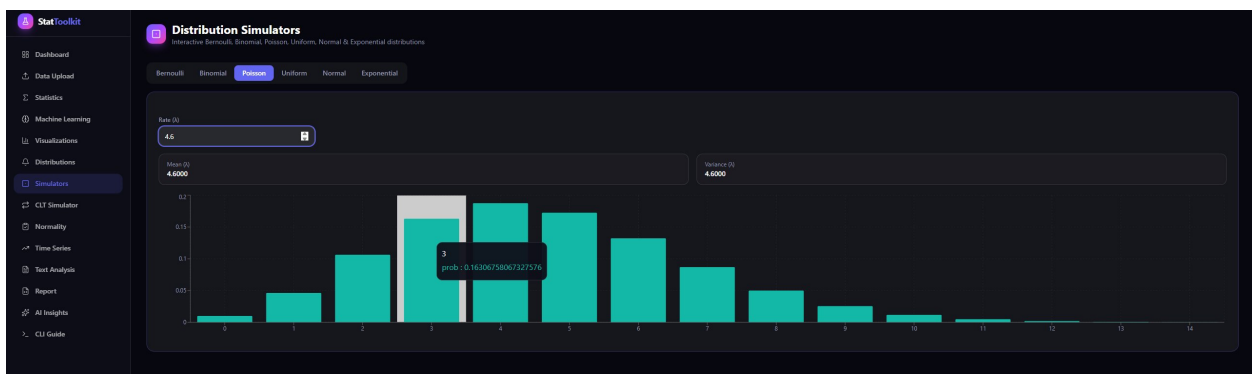


Figure 8 — Distribution Simulators: Poisson Distribution

Interactive Poisson simulator with rate $\lambda=4.6$, matching the approximate average goals per match in the dataset. The chart displays PMF probabilities across $k=0$ to $k=14$ events, with the tooltip showing $P(X=3) \approx 0.163$.

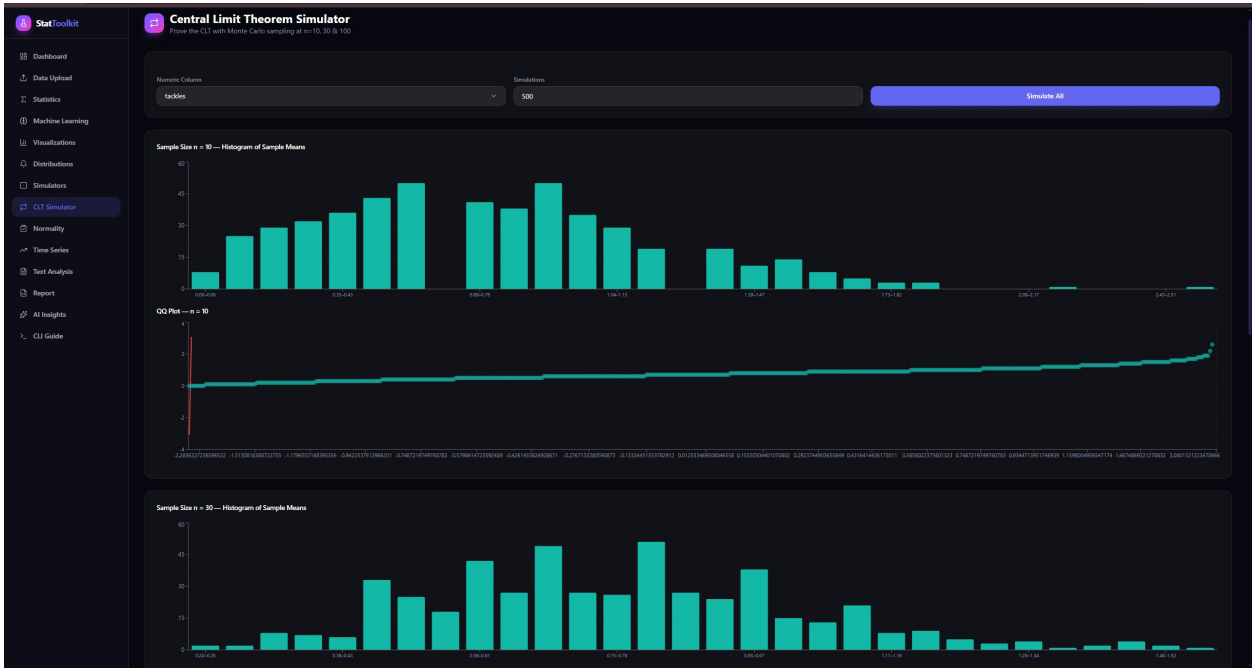


Figure 9 — CLT Simulator: Monte Carlo Sampling (n=10 and n=30)

Central Limit Theorem proof via Monte Carlo simulation (500 iterations) on the tackles column. At $n=10$ the histogram of sample means is noticeably right-skewed; by $n=30$ the distribution visibly converges toward a symmetric bell shape, demonstrating asymptotic normality.

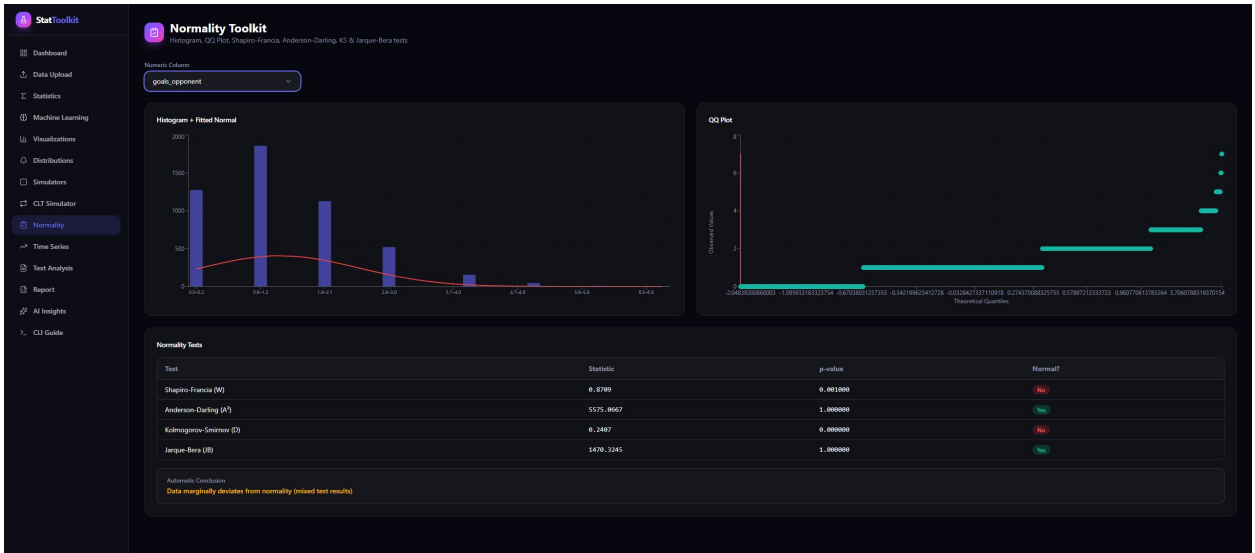


Figure 10 — Normality Toolkit: Histogram, QQ Plot & Formal Tests

Normality evaluation for goals_opponent. The right-skewed histogram and curved QQ plot indicate non-normality. Formal tests confirm this: Shapiro-Francia ($p=0.001$) and Kolmogorov-Smirnov ($p=0.000$) reject H_0 , while Anderson-Darling and Jarque-Bera return $p=1.000$ (mixed results). Conclusion: data marginally deviates from normality.



Figure 11 — Time Series Analysis: Linear Trend Forecasting

Linear trend forecasting applied to the red_cards column over the match time index. The actual values (binary spikes at 0 or 1) are plotted against a flat near-zero linear trend (slope ≈ 0.00045), confirming that red-card occurrences show no significant upward or downward temporal trend across the tournament.

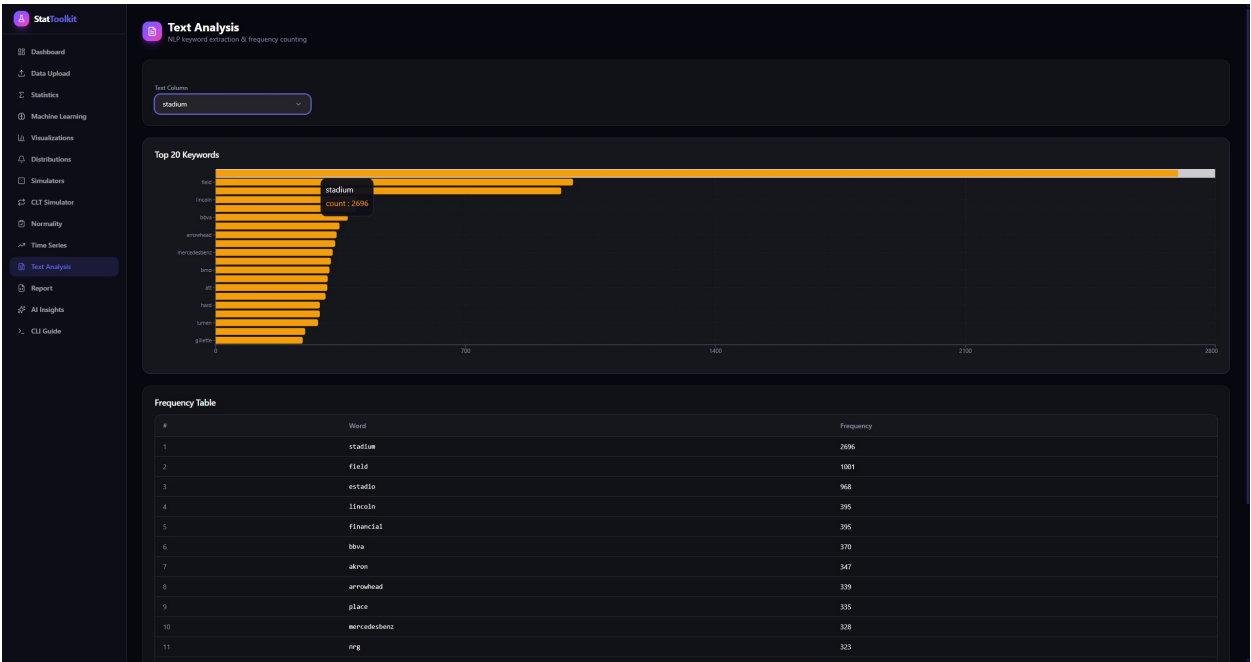


Figure 12 — AI Insights: Automated Anomaly Detection

The AI Insights engine flags all columns with observations exceeding 3σ from the column mean. Notable findings include 260+ goal outliers, 247 assist outliers, 117 market_value_eur outliers, and 206 shots_on_target outliers — identifying statistically exceptional match performances across the dataset.

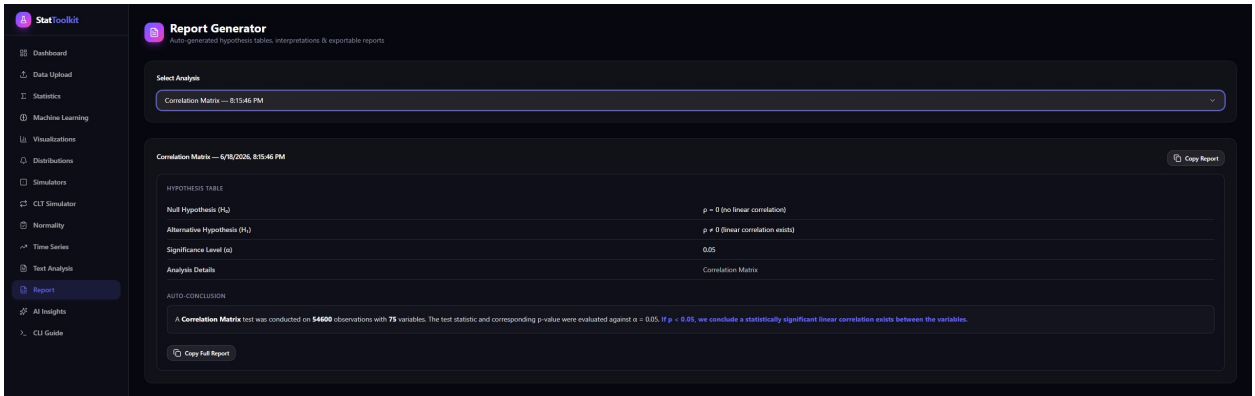


Figure 13 — Report Generator: Auto-generated Hypothesis Table

The Report Generator exports a structured hypothesis summary for the Correlation Matrix analysis. It states H_0 ($\rho=0$, no linear correlation) vs H_1 ($\rho\neq0$), sets $\alpha=0.05$, and auto-generates a plain-language conclusion. The report was run on 54,600 observations across 75 variables.

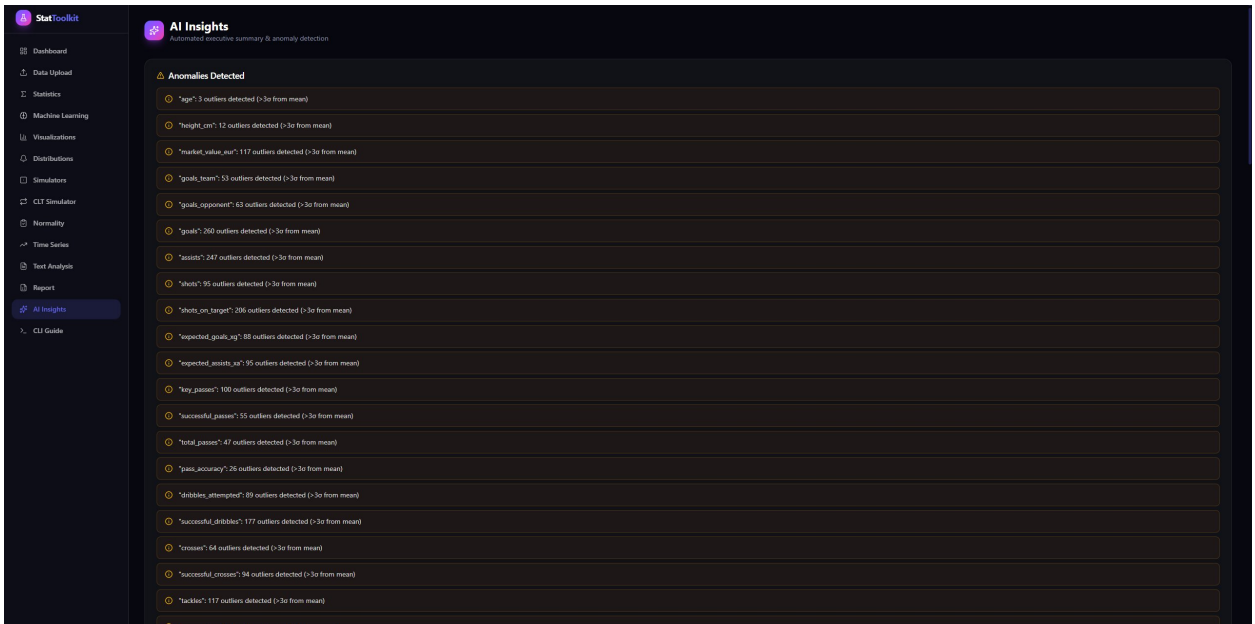


Figure 14 — Text Analysis: NLP Keyword Frequency (stadium column)

NLP tokenization and keyword frequency analysis applied to the stadium text column. The top-20 horizontal bar chart reveals 'stadium' (2,696), 'field' (1,001), and 'estadio' (968) as the most frequent tokens, reflecting the multilingual venue naming conventions across the 2026 World Cup host cities.

